

Sustainable Food Systems and Nutrition:

A Lecture

A food system encompasses everything involved in feeding a population — from growing, harvesting, and processing food, to distributing, consuming, and disposing of it. A sustainable food system meets current nutritional needs without compromising the ability of future generations to meet theirs. It balances three pillars: environmental health, economic viability, and social equity.

MACRONUTRIENTS AND THEIR ROLES

The human body requires three macronutrients in large quantities. Proteins are composed of amino acids, of which 9 are essential — meaning the body cannot synthesize them and must obtain them through diet. Proteins are critical for tissue repair, enzyme production, and immune function. Carbohydrates are the body's primary and preferred energy source; they are broken down into glucose, which fuels cellular respiration and ATP production. Dietary fibre, a type of complex carbohydrate, is indigestible but feeds gut microbiota and regulates bowel function. Fats serve as long-term energy stores, support cell membrane integrity, and are required for the absorption of fat-soluble vitamins A, D, E, and K.

THE FOOD ENVIRONMENT AND BIODIVERSITY

Soil health is foundational to sustainable agriculture. Healthy soil contains organic matter, beneficial microorganisms, and adequate mineral content. The practice of monoculture — growing a single crop repeatedly on the same land — degrades soil, reduces biodiversity, and increases vulnerability to pests and disease. Crop rotation and intercropping are strategies that restore nutrients and support

biodiversity. Agricultural biodiversity, meaning the variety of crops and livestock breeds in use, is essential for food system resilience. Loss of biodiversity narrows the genetic base of our food supply, increasing the risk of large-scale crop failure.

FOOD SECURITY

Food security exists when all people, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food. The four pillars of food security are: Availability (food is produced or imported in sufficient quantities), Access (people can afford and physically reach food), Utilization (the body can absorb and use nutrients from food effectively), and Stability (access is consistent over time, not disrupted by shocks like drought or conflict). Approximately 733 million people face hunger globally, driven by poverty, conflict, climate change, and inequality — not a shortage of total food production.

THE NITROGEN CYCLE AND FOOD PRODUCTION

Nitrogen is essential for plant growth, forming the backbone of amino acids and nucleic acids. The nitrogen cycle converts atmospheric nitrogen into forms usable by plants, primarily through nitrogen fixation by soil bacteria such as *Rhizobium*, which form symbiotic relationships with legume roots. Synthetic nitrogen fertilizers, developed via the Haber-Bosch process, have dramatically increased crop yields but also cause significant environmental harm. Excess nitrogen runs off into waterways, causing eutrophication — algal blooms that deplete oxygen and kill aquatic life. This is a major negative externality of industrial agriculture.

FOOD PROCESSING AND THE NOVA CLASSIFICATION

Not all food processing is harmful. The NOVA classification system groups foods into four categories based on the extent and purpose of processing. Group 1 consists of unprocessed or minimally processed foods such as fruits, vegetables, eggs, and plain meat. Group 2 includes processed culinary ingredients like oils, flour, and sugar. Group 3 covers processed foods such as canned vegetables or artisan cheese. Group 4, ultra-processed foods, includes products formulated from industrial ingredients with additives for flavour, colour, and shelf life — such as packaged snacks, soft drinks, and reconstituted meat products. High consumption of Group 4 foods is strongly associated with obesity, type 2 diabetes, cardiovascular disease, and poor diet quality.

GREENHOUSE GAS EMISSIONS AND FOOD

The global food system is responsible for approximately 26 to 34 percent of total greenhouse gas emissions. Livestock farming is the single largest contributor within the food sector, accounting for roughly 14.5 percent of global emissions, primarily through methane from enteric fermentation in ruminants and nitrous oxide from manure. Red meat — especially beef — has the highest carbon footprint per kilogram of protein. Plant-based diets have significantly lower emissions; shifting consumption toward legumes, whole grains, fruits, and vegetables is considered one of the most impactful individual actions for reducing food-related emissions. Food loss and waste account for around 8 to 10 percent of global greenhouse gas emissions.